FPGA-Based Architecture for a Multisensory Barrier to Enhance Railway Safety

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***Abstract*— Due to the continuous development of the high-speed railway lines and the growth of automatic systems, it is becoming more and more necessary to install safety elements to prevent accidents. One of these elements is the so-called obstacle fall detector on the tracks, in charge of detecting the sudden appearance of dangerous objects in the safety area of the tracks. For that purpose, this paper proposes the design of a multisensory system, consisting of two barriers, one emitting and another receiving, which are placed at opposing sides of the railway tracks, respectively, and they use infrared (IR) and ultrasonic (US) sensors in order to establish optical and acoustic links between them. From the processing point of view, this proposed system can be divided into the low and high levels. Whereas the high-level stage corresponds to the monitoring system, providing information about the state of the tracks (free or occupied by obstacles), the low-level one is responsible for managing emissions and detections carried out by the sensory system. The work presented here is focused on the design and implementation of the low-level stage in a field-programmable gate array device, based on the encoding of the sensory signals for both the US and the IR one. The experimental results have successfully validated the feasibility of the proposal, providing a high reliability to the detection system.**

***Index Terms*— Efficient encoding schemes, field-programmable gate array (FPGA)-based implementation, obstacle fall detection, railway safety.**

I. Introduction

MODERN transport systems are designed under the principles of safety and reliability, and the development of high-speed railway lines is based on such premises [1], [2]. In fact, the projects that are currently being developed [3] show the constant need for improvement in railway safety. Some of the circumstances that receive relevant attention are the quality of the tracks [4], [5] and the existence of objects on them [6]. This work is focused on the latter.

In the case of standard railway lines, a large number of dedicated sensory systems have been installed in areas at or near level crossings to prevent collisions between trains and vehicles [7]. On high-speed lines, there are no level crossings, but zones close to bridges or tunnels are considered to be quite critical, since objects can fall onto the tracks. This can be caused by the fall of a vehicle or any material transported by a vehicle onto the line. Landslides can also happen at the entrances and exits of tunnels. In these critical areas, if there is a system to detect the presence of obstacles, railway traffic can be halted and possible accidents can be avoided. According to the Spanish Railway Regulations [8], a system is required to detect obstacles in such areas in high-speed lines. This system has to be able to detect objects larger than $50 \times 50 \times 50$ cm^3, scanning the safety area every 500 ms. The regulations indicate that the system has to be an infrared (IR) barrier, consisting of emitter–receiver pairs, each placed at opposing sides of the line. García *et al.* [9], [10] have conducted a research project in this field and have proposed a detection system, which not only meets the requirements of the aforementioned regulations but also improves the reliability of the detection system.

Reliability is highly influenced by the design of the sensor used, the conditions in which the sensor is working, and the signal processing that is carried out by the system. In some adverse conditions of a railway environment, there are often significant differences in the performance from various types of sensors, as every type considered has its own inherent weaknesses and limitations. For example, weather conditions (rain, fog, etc.) are of critical importance when using cameras [11]; optical sensors produce false alarms due to solar radiation or weather conditions; the same problem is found using ultrasonic (US) sensors if the system is working in a turbulent atmosphere (e.g., when wind speeds are high or fluctuating) [12]. Since the ideal sensor does not exist, Trankler and Kanoun [13] and Bai *et al.* [14] have developed a multisensory barrier consisting of IR and US sensors, so that the drawbacks of using any particular type of sensor are compensated for by the performance characteristics from the other type, and vice versa. For example, it is very common to get detection problems with IR sensors in foggy days, whereas there is no problem with the US ones. Conversely, in windy days, such detection problems can appear with the US sensors but not with the IR ones. García *et al.* [10] show the complementarity of these sensors and how the detection problems of each one are compensated for.

Apart from the use of a multisensory barrier, there are other factors that improve the reliability of the detection system [10].

1) The use of a robust encoding scheme that simultaneously provides multiple emissions and receptions, avoids

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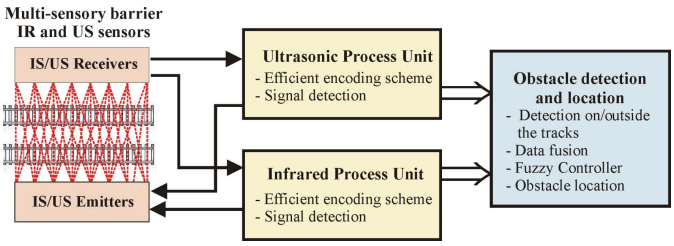


Fig. 1. Block diagram of the developed global detection system.

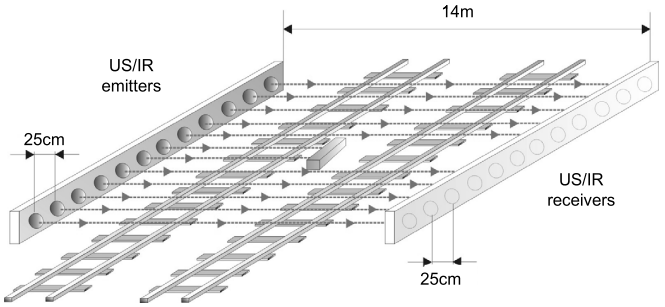


Fig. 2. Geometrical distribution of the multisensory barrier.

crosstalk interference among emissions, and can work with a low signal-to-noise ratio (SNR).

2) Data fusion techniques applied to the merging of information from both sensors (IR and US) can also enhance the system performance.

Some functional aspects included in the proposed design of the obstacle detection system imply an added value compared with the requirements stated in [8], namely, obstacle location on the tracks or outside the tracks, discrimination between mobile and fixed objects, estimation of the object size, and reduction of false alarms. False alarms generate financial losses whenever the system detects an obstacle that does not exist. Optimal estimation techniques based on the Kalman filter and $H\infty$ filtering have been already used to minimize the generation of false alarms [9], [15]. García *et al.* [9], [15] proposed two processing levels for the detection system: the low level, which includes the transmission and detection of the encoded emissions, and the high level, concerning the monitoring system located at the control center.

As shown previously, García *et al.* [9], [15] have already carried out previous research related to the development of the multisensory barrier and the computer-based algorithms for improving its reliability. This particular work is focused on the design of an efficient and low-cost field-programmable gate array (FPGA)-based architecture for the implementation of the low-level processing associated with a US and IR multisensory barrier dedicated to the obstacle detection in high-speed railway lines. The proposed sensory processing, based on signal encoding, has been optimized for the corresponding implementation, thus allowing a real-time performance to be achieved, since the processing time ($\approx$77 ms) is lower than the maximum allowed scanning time (500 ms according to [8]). The experimental results with the developed prototype have successfully validated the design. The rest of this paper is organized as follows. Section II describes the structure of the multisensory system. Section III deals with the signal processing for every sensor, including the signal emission and detection. Section IV describes the proposed architecture for the implementation of the low-level processing. Section V shows some experimental results obtained with the implemented system. Finally, conclusions are discussed in Section VI.

II. Global Overview

Fig. 1 shows the developed system for obstacle detection, widely described in [9] and [10]. It is divided into three stages: 1) sensory modules based on IR and US transducers; 2) the process units (low-level processing); and 3) the obstacle detection and location module (the monitoring system).

The multisensory barrier is composed of two IR and US barriers, one continuously emitting and the other receiving, placed at both sides of the railway tracks, as is shown in Fig. 2. This scheme provides a net of links among emitters and receivers. Due to the fact that the minimum dimensions of the object to be detected are $50 \times 50 \times 50$ cm^3, the distance between continuous transducers has been fixed at 25 cm. In this way, if an object with minimum dimensions is in the scanned area, at least two links of each technology (IR and US) are interrupted. On the other hand, the distance between emitters and receivers is 14 m, given by the width of the railroads, although it could be longer. Fig. 2 shows the described geometry of the detection system as well.

García *et al.* [9], [10] have described the multisensory barrier and the obstacle detection and location module, the consecutive sections are focused only on the definition of the hardware architecture for the US and IR process units, including efficient encoding schemes and the suitable signal processing. For the sake of clarity, it is considered hereinafter a segment of the multisensory barrier, composed of four IR and four US emitters and eight receivers of each technology, and the corresponding process units. We propose to implement the whole low-level stage in FPGA devices. This fact is really important from the maintenance point of view, as it minimizes the number of elements that can provoke faults, apart from the reliability that FPGAs guarantee in railway applications [16], [17]. Furthermore, FPGAs provide interesting features and adaptability for the implementation of these kinds of applications, where there are numerous parallel data paths, all of them with intensive computing. On the other hand, those approaches based on general-purpose microprocessors have not been considered, since they often present a more limited parallelism and adaptation to the application requirements. Fig. 3 shows the prototype scheme implemented in this paper.

III. Signal Processing

According to the geometry of the system in Fig. 2, the radiation coming from several emitters reaches every receiver. The proposal uses direct sequence code multiple division access techniques [18] to encode the emissions not only for distinguishing them but also for avoiding interferences among emissions of the same nature. The detection of the emissions

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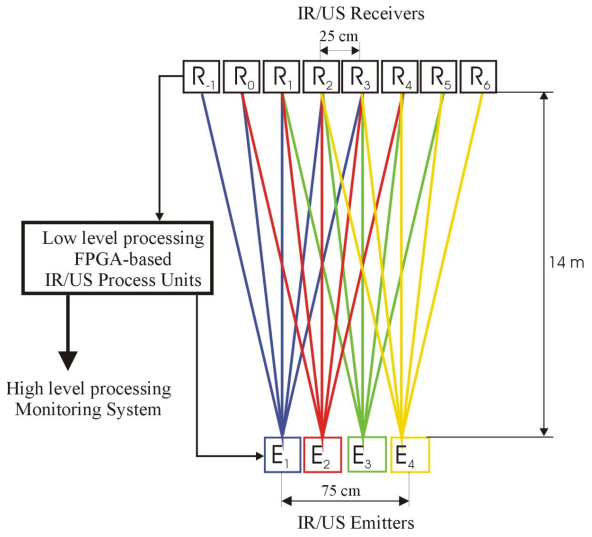


Fig. 3. Scheme of the implemented multisensory barrier.

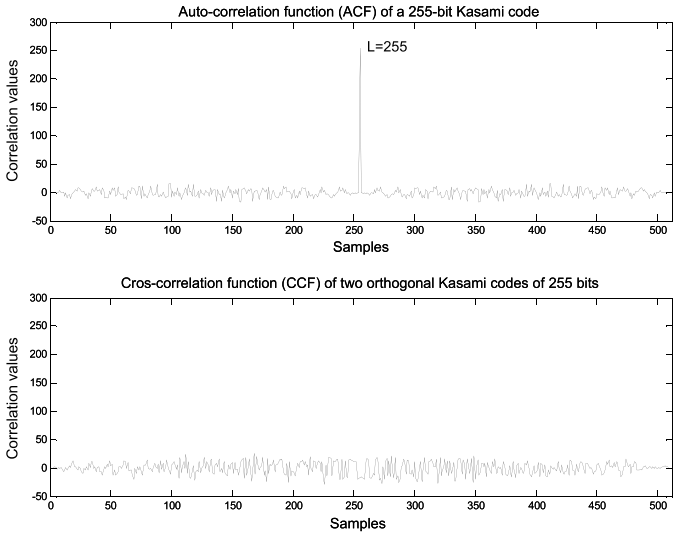


Fig. 4. Example of an ACF and a CCF for Kasami codes of $L^{(\mathrm{us}} = 255$ b.

is performed using correlation techniques. Every receiver correlates the receiving signal with the codes assigned to be detected (the ones belonging to the expected emissions). The process gain derived from the use of correlation techniques provides higher immunity to noise and guarantees the barrier operation even under adverse environmental conditions.

Due to the fact that two different physical sensors are used, it is necessary to define proper signal processing for each one of them, choosing suitable encoding and modulation schemes. After comparing different encoding approaches [19], Kasami sequences [20] are used for the US emissions and complementary sets of sequences (CSSs) [21] for the IR ones. Both strategies guarantee a high autocorrelation function (ACF), so the emissions can be easily detected, and a reduced cross-correlation function (CCF) among sequences, in order to mitigate interferences in this multiple-source emission scheme. The proposed US sensory is not coherent, where Kasami sequences have already shown a better performance than other pseudorandom codes or CSS [22]. On the other hand, the proposed IR system is a coherent link, and CSS can be applied since they provide a perfect ACF and a CCF in those conditions [21], [23]. Furthermore, CSS present efficient correlation architectures [23].

The parameters and variables involved in each signal processing share similar names (e.g., code length L and samples per carrier period m). In order to distinguish them, we use a superscript for each type of signal: US, $^{(\mathrm{us}}$, and IR, $^{(\mathrm{ir}}$.

A. Ultrasonic Sensors

As has been previously mentioned, Kasami sequences have been selected for the US emissions because they provide suitable correlation bounds in systems with several emitters and receivers working simultaneously. In [24], these sequences were already used successfully in a sensory system for a track circuit, based on electrical transmissions through the railway tracks. In this case, the signal encoding can be considered similar to the one proposed in [24], although it consists of US transmissions with different operation frequencies, bandwidths, and channel constraints. The periodic ACF$[n]$ and the periodic CCF$[n]$ for a Kasami sequence $c_i^{(\mathrm{us}}$ can be observed in

$$\mathrm{ACF}[n] = \begin{cases} L^{(\mathrm{us} & n = 0 \\ \{-1, -2^{l/2} - 1, 2^{l/2} - 1\} & n \neq 0 \end{cases} \quad (1)$$

$$\mathrm{CCF}[n] = \{-1, -2^{l/2} - 1, 2^{l/2} - 1\} \quad \forall n \quad (2)$$

where $L^{(\mathrm{us} = 2^l - 1$ is the length of the sequences, $2^{l/2}$ is the number of available sequences, and l is the even number of bits of the linear feedback shift register used to generate the Kasami code. Fig. 4 shows an example of an aperiodic ACF and CCF for Kasami codes of 255 b. Note that in an aperiodic correlation, sidelobes have values different from those described in (1) and (2). The length of 255 b has been determined as a tradeoff between a suitable process gain (for low SNRs in transmissions) and a reduced computational load. Nevertheless, this length could be changed in order to adapt the proposal to different environmental conditions.

The signal $e_j^{(\mathrm{us}}[n]$ to be transmitted is obtained by binary phase shift keying (BPSK) modulating the Kasami sequence $c_j^{(\mathrm{us}}$, as shown in

$$e_j^{(\mathrm{us}}[n] = \sum_{l=0}^{L^{(\mathrm{us}}-1} c_j^{(\mathrm{us}}[l] \cdot s^{(\mathrm{us}}[n - l \cdot m^{(\mathrm{us}}] \quad (3)$$

where $e_j^{(\mathrm{us}}[n]$ is the modulated signal transmitted by the US emitter j, $L^{(\mathrm{us}}$ is the length of the code $c_j^{(\mathrm{us}}$ used in the encoding, $s^{(\mathrm{us}}[n]$ is the carrier period used in the modulation, and $m^{(\mathrm{us}}$ is the number of samples per carrier period.

With regard to the reception module, it can be divided into two main stages. First, the signal received by the transducer i is BPSK demodulated, according to

$$d_i^{(\mathrm{us}}[n] = \sum_{l=0}^{m^{(\mathrm{us}}-1} r_i^{(\mathrm{us}}[l + n] \cdot s^{(\mathrm{us}}[l] \quad (4)$$

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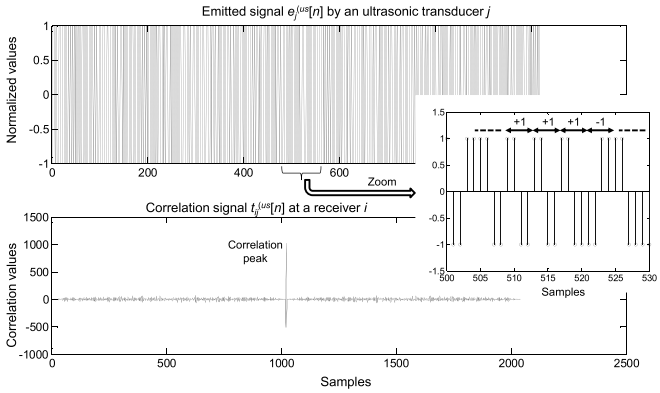


Fig. 5. Example of US encoding for a 255-b Kasami code with a carrier frequency of $f_c^{(\mathrm{us}} = 50$ kHz and a sampling one of $f_S^{(\mathrm{us}} = 200$ kHz.

where $d_i^{(\mathrm{us}}[n]$ is the output from the demodulation stage, $r_i^{(\mathrm{us}}[n]$ is the input signal coming from the acquisition stage, and $s^{(\mathrm{us}}[n]$ is the symbol of the demodulation, represented by $m^{(\mathrm{us}}$ samples.

Second, the demodulated output $d_i^{(\mathrm{us}}[n]$ has to be correlated in order to search for possible emissions. Since every emitter has a different code assigned, every receiver has to implement as many correlators as emissions have to be detected. This correlation process is described in

$$t_{ij}^{(\mathrm{us}}[n] = \sum_{l=0}^{L^{(\mathrm{us}}-1} d_i^{(\mathrm{us}}[l \cdot m^{(\mathrm{us}} + n] \cdot c_j^{(\mathrm{us}}[l] \tag{5}$$

where $d_i^{(\mathrm{us}}[n]$ is the demodulated signal, $c_j^{(\mathrm{us}}[n]$ is the code emitted by a transducer j, $L^{(\mathrm{us}}$ is the length of the mentioned code, and $t_{ij}^{(\mathrm{us}}[n]$ is the correlation output, corresponding to the receiver i searching for the emission j. Fig. 5 shows this encoding processing for the US transducers with a carrier frequency $f_c^{(\mathrm{us}}$ of 50 kHz and a sampling frequency $f_S^{(\mathrm{us}}$ of 200 kHz. Sidelobes appear in the correlation signal $t_{ij}^{(\mathrm{us}}[n]$ due to noncoherent detection. An analysis of the performance of Kasami codes, BPSK modulated for US emissions, can be found in [25].

B. Infrared Sensors

As was introduced previously, the IR emissions have been encoded by mutually orthogonal complementary sets of sequences. Let ϕ_{SiSi} denote the autocorrelation function of the binary sequence $S_{i,M}$, and let $\phi_{\mathrm{SiSi}}[n]$ denote the nth element in the function ϕ_{SiSi}. A set of M binary sequences $\{S_{i,M},\ 1 \le i \le M\}$ with length L, whose elements are either $+1$ or -1, is a complementary set if the sum of their aperiodic autocorrelation functions is zero for all nonzero shifts, and it meets

$$\sum_{i=1}^{M} \phi_{\mathrm{SiSi}}[n] = \begin{cases} M \cdot L & \text{if } l = 0 \\ 0 & \forall\ l \neq 0. \end{cases} \tag{6}$$

Let $\phi_{\mathrm{SiS'i}}$ denote the cross-correlation function of the complementary sequences $S_{i,M}$ and $S'_{i,M}$, and let $\phi_{\mathrm{SiS'i}}(n)$ denote

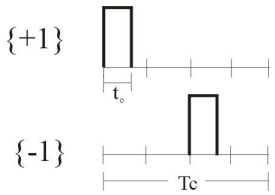


Fig. 6. PPM excitation for IR emitters and symbol assignment.

the nth element in the function $\phi_{\mathrm{SiS'i}}$. The sets of sequences $(S_{i,M}, S'_{i,M} 1 \le i \le M)$ with length L are mutually orthogonal if it meets

$$\sum_{i=1}^{M} \phi_{\mathrm{SiS'i}}[n] = 0 \quad \forall l. \tag{7}$$

As Fig. 3 depicts, a prototype of the barrier is designed with four IR emitters, so a set of $M = 4$ sequences is needed to encode every IR emitter. Nevertheless, this number can be increased according to the properties of CSS ($L = M^N$ and $M = 2^p$, with p and N any positive integer), without limiting the generality of the foregoing description. The transmission of the $M = 4$ sequences $S_{i,M}^j$ in a set through the IR emitter j is accomplished by first interleaving these sequences to generate a new sequence $c_j^{(\mathrm{in}}$ with length $L^{(\mathrm{in}} = 4 \cdot L$, defined as

$$\begin{aligned} c_j^{(\mathrm{in}} = {} & S_{1,4}^j[1] \quad S_{2,4}^j[1] \quad S_{3,4}^j[1] \quad S_{4,4}^j[1] \\ & \ldots S_{1,4}^j[L] \quad S_{2,4}^j[L] \quad S_{3,4}^j[L] \quad S_{4,4}^j[L]. \end{aligned} \tag{8}$$

The resulting sequence $c_j^{(\mathrm{in}}$ is then modulated according to pulse position modulation (PPM). It means that a pulse of duration t_o is transmitted for each bit $c_j^{(\mathrm{in}}$ $\{+1, -1\}$. Depending on the time position of that pulse in the period of the emitted signal T_c, the bit is either $+1$ or -1. Fig. 6 shows the pulse assignment for every bit in the proposed PPM modulation. Note that the maximum efficiency of IR emitters is achieved for pulsed emissions.

In this way, the signal $e_j^{(\mathrm{in}}[n]$ to be transmitted by the emitter j is the result of modulating the new interleaved sequence $c_j^{(\mathrm{in}}$ with the corresponding PPM pulse in Fig. 6. As the sequence $c_j^{(\mathrm{in}}$ is the interleaved version of $M = 4$ individual sequences, it is necessary to carry out the following process in the reception module in order to identify the corresponding emissions: 1) PPM demodulation to obtain $d_i^{(\mathrm{in}}[n]$; 2) to undo the interleaving; 3) to perform the four correlation functions; and 4) the final sum of the correlation functions to get $t_{ij}^{(\mathrm{in}}[n]$, the signal detected by the receiver i searching for the emission j. Fig. 7 summarizes the whole process and Fig. 8 depicts the received and detected signals when $M = 4$, $L^{(\mathrm{in}} = 64$, and SNR $= 0$ dB. For a continuous emission and detection, the ideal peak detector output $\mathrm{PD}_{ij}[n]$ in Fig. 7 meets

$$\mathrm{PD}_{ij}[n] = M \cdot L \sum_{k=-\infty}^{k=\infty} \delta[n - k \cdot L]. \tag{9}$$

The robustness of the detection system using both encoding schemes has been previously analyzed through simulations and

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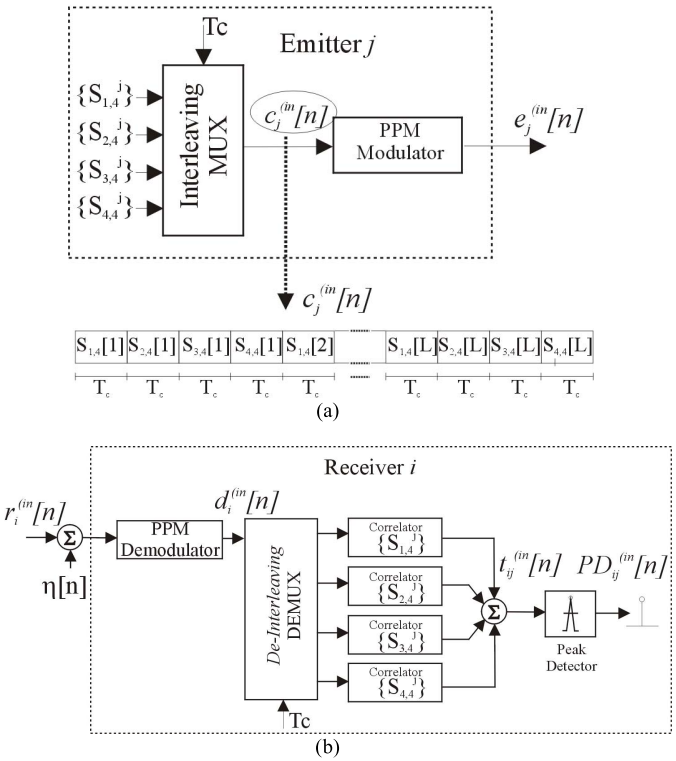


Fig. 7. Proposed IR signal processing for the emitting and receiving stages. (a) Emitter j. (b) Receiver i.

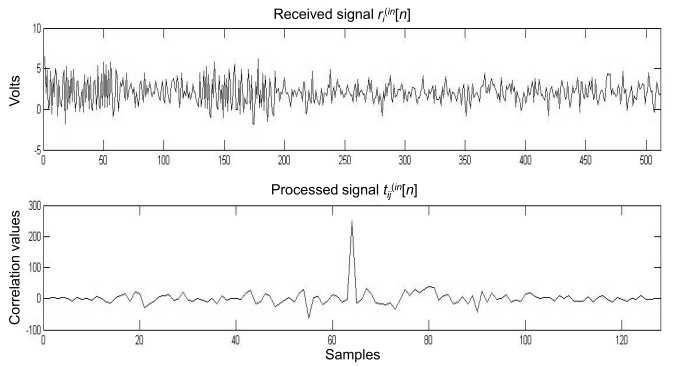


Fig. 8. Example of one IR detection for four-CSS, when $M = 4$, $L^{(\text{in}} = 64$, and SNR $= 0$ dB: received and processed signals are $r_i^{(\text{in}}[n]$ and $t_{ij}^{(\text{in}}[n]$, respectively.

some real tests, covering different weather conditions (clear days, dense fog, and changeable wind speed) and different noise levels [9], [10]. These previous tests show that the crosstalk between channels is very low (compared with the correlation peaks), so it does not affect to the detection system. Furthermore, the obtained results confirm a correct working even with SNR $= -6$ dB (very unusual for this particular application).

IV. Proposed Architecture

The architecture presented here can be divided into three different blocks, according to the three FPGA-based platforms used for the implementation. As can be observed in Fig. 3, the eight emitters (four US and other four IR) are controlled from only one FPGA-based platform, which provides the driving signals to the transducers, as well as the emission synchronization necessary for the IR receiver. On the other hand, the processing modules for the eight US receivers have been included in another FPGA-based platform, whereas the processing for the eight IR receivers has been implemented on a third FPGA-based board.

A. Ultrasonic and Infrared Emitters

In this emission module, three main blocks can be mentioned.

1) There is a clock manager, capable of generating the clock enable signals for the US emitters and the IR emitters. Furthermore, this block also provides the emission synchronization that starts the emission and reception processes in the whole system.
2) There are some blocks related to the US emitter.
3) It is possible to find similar blocks for the IR emission.

Fig. 9 shows a global block diagram of the emission module, where these three parts can be identified.

The clock manager provides three different clock enables. The 200-kHz clock enable is used by the US emitters to carry out the corresponding BPSK modulation, so the final signal $e_i^{(\text{us}}[n]$ emitted by a transducer i has a carrier of $f_c^{(\text{us}} = 50$ kHz, with an oversampling $m^{(\text{us}} = f_e^{(\text{us}}/f_c^{(\text{us}} = 200$ kHz/50 kHz $= 4$. On the other hand, the 100-kHz clock enable is used by the IR blocks to perform the PPM modulation with a period of 100 μs. Finally, a synchronization and *init* signal with a frequency of 13 Hz is provided to simultaneously fire all the emitters and synchronize the IR receivers (note that the US receivers do not require any synchronization with the emission).

With regard to the emitters, both US and IR, they have a similar internal structure, where it is possible to find a modulation (BPSK for US and PPM for IR signals) for each transducer. Every two modulated signals, $e_j^{(\text{us}}[n]$ and $e_j^{(\text{in}}[n]$, are processed by a digital-to-analog converter (DAC) controller, which generates the control and input signals for the DAC used: DAC121S101 [26]. The system includes four converters and their corresponding controllers.

Fig. 10 shows the diagram for the BPSK modulator of the US emitters. The Kasami code $c_i^{(\text{us}}$ to be emitted (with a length $L^{(\text{us}})$ as well as a period $s^{(\text{us}}[n]$ of the carrier for the modulation has been stored in both RAM memories. Every time the emission is fired, the address manager starts to provide address values for both memories, and it increases them so that the whole carrier period is read before a new Kasami bit is obtained from the corresponding memory. The BPSK modulation is carried out by means of a XNOR gate, as can be observed in Fig. 10. According to the system features commented in the previous sections, the memory sizes should be $L^{(\text{us}} \times 1 = 255 \times 1$ b and $m^{(\text{us}} \times 1 = 4 \times 1$ b for the Kasami code and the carrier period, respectively.

On the other hand, the IR emission has been encoded by four-CSS with a length of $L^{(\text{in}} = 64$ b. These sequences have been interleaved in order to form a single sequence

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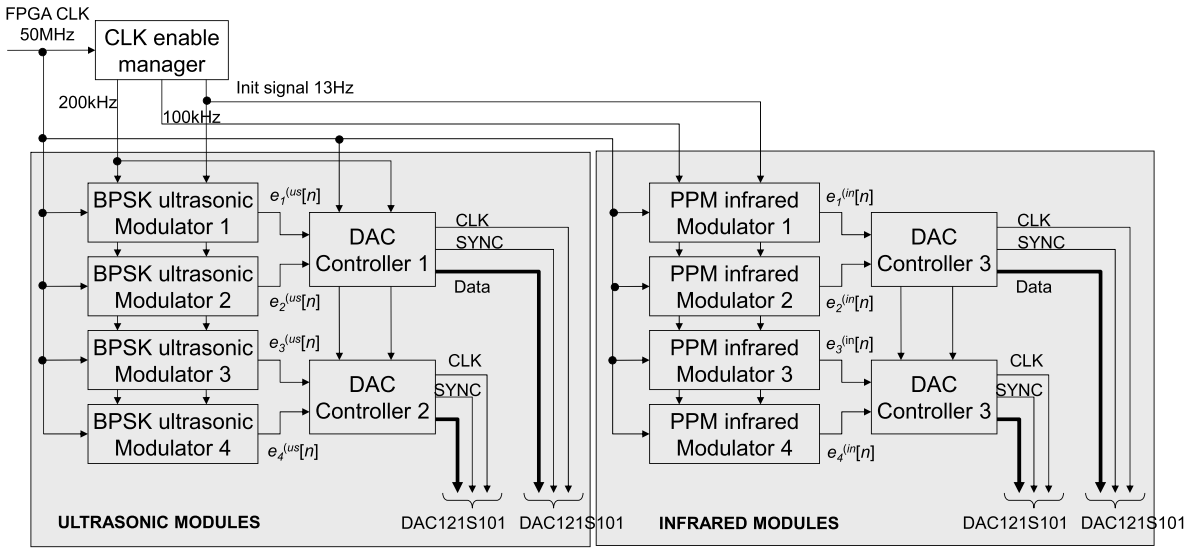


Fig. 9. Block diagram of the emission module.

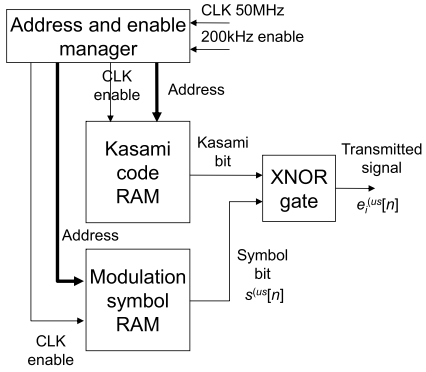


Fig. 10. Block diagram of the BPSK modulation for US transducers.

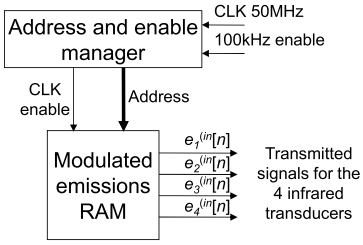


Fig. 11. Block diagram of the modulation for IR transducers.

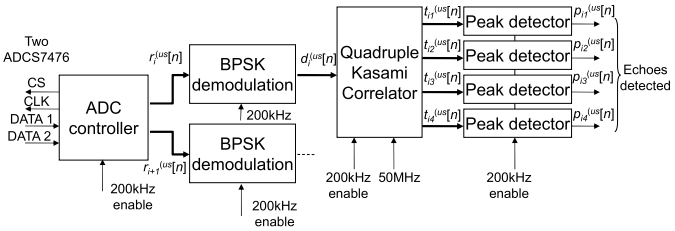


Fig. 12. Block diagram of the US receiving block.

to be transmitted. The final signal $e_j^{(\text{in}}[n]$ to be emitted by every IR transducer j has been generated offline, considering the proposed PPM modulation, and they have been stored in a RAM memory. Again, after every emission *init* signal, a controller starts to provide the corresponding address for the memory, so the signals $e_j^{(\text{in}}[n]$ of the four transducers are read simultaneously. Fig. 11 describes the mentioned design. Since the carrier frequency of the PPM modulation is $f_c^{(\text{in}} = 10$ kHz and the frequency of the emitter is $f_e^{(\text{in}} = 100$ kHz, the memory size should be $4 \cdot L^{(\text{in}} \cdot m^{(\text{in}} \times 4 = 4 \cdot L^{(\text{in}} \cdot f_e^{(\text{in}}/f_c^{(\text{in}} \times 4 = 4 \cdot 64 \cdot 100$ kHz/10 kHz $\times 4 = 2560 \times 4$ b.

B. Ultrasonic Receivers

The reception module of the US transducers is based on two main computations: the BPSK demodulation and the correlation searching for the emitted Kasami codes $c_i^{(\text{us}}$. As can be observed in Fig. 12, the proposal consists of an ADC controller, in charge of communicating and managing the acquisition channels available in the two ADCS7476 devices (two channel per device), and a demodulation block, which processes the acquisition $r_i^{(\text{us}}[n]$ from a channel i to provide a signal $d_i^{(\text{us}}[n]$ where up to four Kasami codes simultaneously emitted can be searched by means of a quadruple correlator.

Finally, the four correlation outputs $t_{ij}^{(\text{us}}[n]$ are analyzed in order to determine local maximum values that can be considered as transmissions, where the index $i = \{1, 2, 3, 4\}$ denotes the receiver and the index $j = \{1, 2, 3, 4\}$ represents the emitter. Note that the ADC is a 12-b converter, where all the bits are considered in the next stages, and the data width is extended if necessary to avoid any overflow.

The whole design is computed at 200 kHz, the sampling frequency $f_S^{(\text{us}}$, although the quadruple correlator requires to be operated at $f_{\text{CLK}} = 50$ MHz to compute the four correlations in real time for every sample coming from the demodulation. Note that since every correlator consists of two parallel accumulators (one per each half of the correlated sequence), a 255-b Kasami code requires $L^{(\text{us}}$ cycles of the

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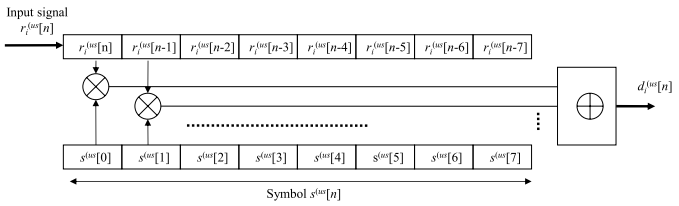


Fig. 13. Block diagram of the BPSK demodulation.

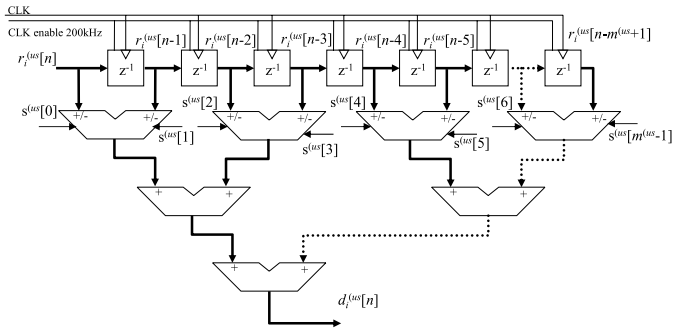


Fig. 14. Architecture proposed for the BPSK demodulation.

50-MHz clock signal to compute the correlation for every input signal $d_i^{(\mathrm{us}}[n]$. The length $L^{(\mathrm{us}}$ of the Kasami codes can be increased without modifying the architecture as far as $f_{\mathrm{CLK}} \geq L^{(\mathrm{us}} \cdot f_S^{(\mathrm{us}}/2$. If necessary, more parallel accumulators can be added to every sequence correlator in order to allow longer codes.

The demodulation block should convolve the input signal $r_i^{(\mathrm{us}}[n]$ from the transducer i with the modulation symbol $s^{(\mathrm{us}}[n]$ (see Fig. 13). Since the symbol $s^{(\mathrm{us}}[n]$ is digital, all the multiplications can be reduced to additions and the demodulation can be implemented as shown in Fig. 14.

With regard to the quadruple correlator, it is based on a buffer with a size $L^{(\mathrm{us}} \cdot m^{(\mathrm{us}}$, where $L^{(\mathrm{us}}$ is the length of the Kasami codes and $m^{(\mathrm{us}}$ is the oversampling of the acquisition system: $m^{(\mathrm{us}} = f_S^{(\mathrm{us}}/f_c^{(\mathrm{us}}$. The samples $d_i^{(\mathrm{us}}[n]$ stored in this buffer are used by four parallel accumulators to compute the correlation process with four corresponding Kasami codes $c_i^{(\mathrm{us}}$ stored in ROM memories, providing the outputs $t_{ij}^{(\mathrm{us}}[n]$, for $j = \{1, 2, 3, 4\}$. Fig. 15 shows the block diagram of the quadruple correlator explained. Note that the top module works at a frequency of $f_S^{(\mathrm{us}} = 200$ kHz, whereas the bottom one operates at the global clock frequency $f_{\mathrm{CLK}} = 50$ MHz, so the $L^{(\mathrm{us}}$ samples of a Kasami code can be computed before the next sample $d_i^{(\mathrm{us}}[n]$ arrives.

Finally, the resulting signals $t_{ij}^{(\mathrm{us}}[n]$ from the correlation are processed in order to determine the local maxima in an analysis window of $F^{(\mathrm{us}}$ samples with values higher than the threshold $U^{(\mathrm{us}}$. Fig. 16 shows the block diagram of this peak detector, based on two comparators to consider the threshold $U^{(\mathrm{us}}$ and a counter to take into account the window $F^{(\mathrm{us}}$.

C. Infrared Receivers

The IR receiver presents a structure very similar to that proposed for the US one. It is also based on a demodulation

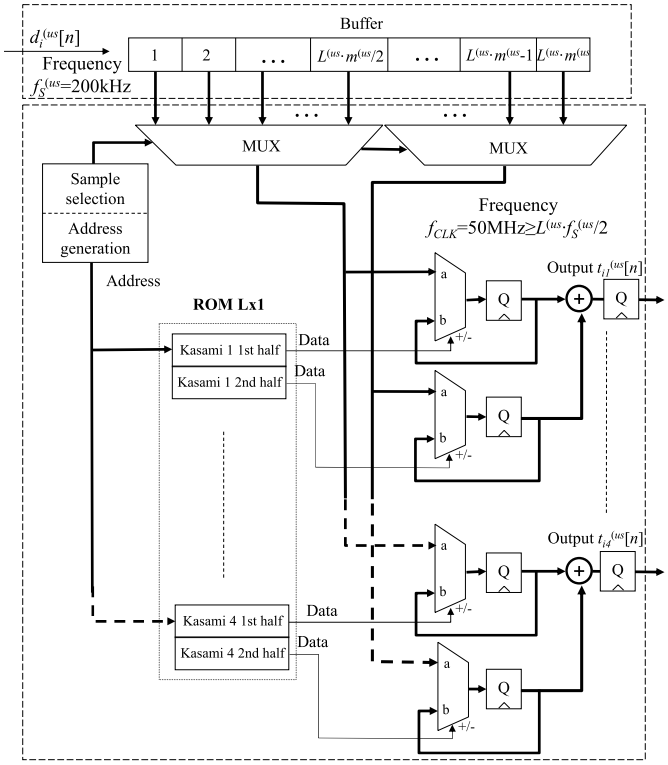


Fig. 15. Block diagram of the quadruple correlator used in the US receiver.

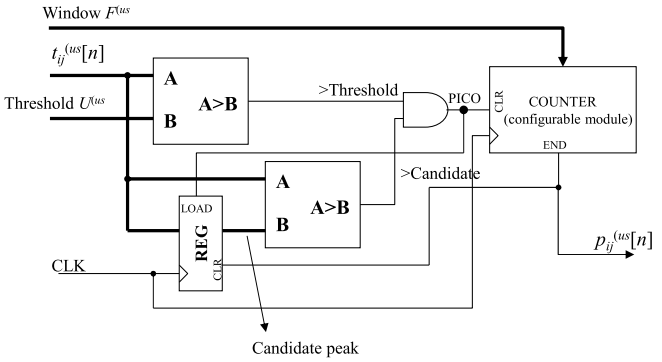


Fig. 16. Block diagram of the peak detector implemented.

block and a correlation process, as can be observed in Fig. 17. The main difference consists in the synchronization with the emission *init*. In this way, it is considered as a synchronous link and it implies a considerable reduction in the computational resources required, since the oversampling $m^{(\mathrm{in}}$ has no influence on the correlation process.

The ADC previously mentioned is also used here to manage the two acquisition channels available in the ADCS7476 device (four devices are used to work with eight IR receivers) [27]. As before, with the US transducers, the 12 b from the ADC are considered and the data width in the next stages is extended to avoid any overflow. The acquired signals $r_i^{(\mathrm{in}}[n]$ are demodulated in the following block, according to a PPM scheme with a carrier frequency of $f_c^{(\mathrm{in}} = 10$ kHz and a sampling frequency of $f_S^{(\mathrm{in}} = 100$ kHz. Fig. 18 depicts the implemented PPM demodulation, where first the current

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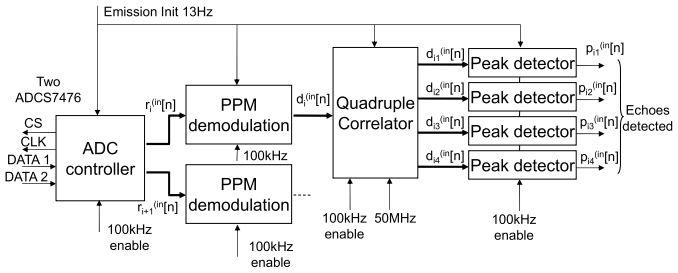


Fig. 17. Block diagram of the IR receivers.

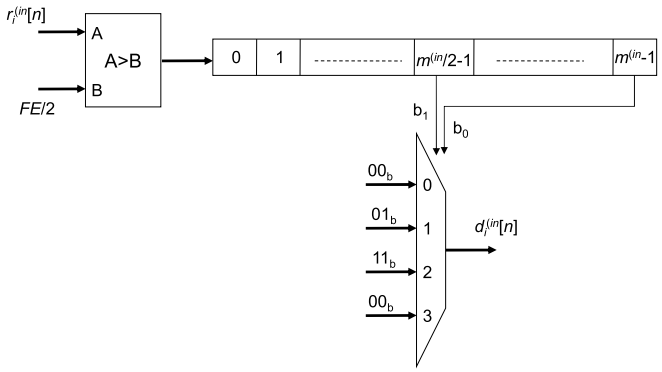


Fig. 18. Block diagram of the PPM demodulation implemented.

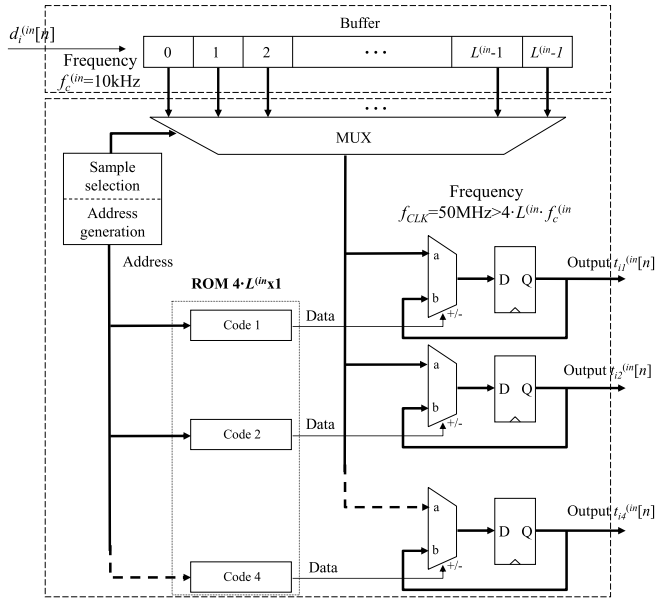
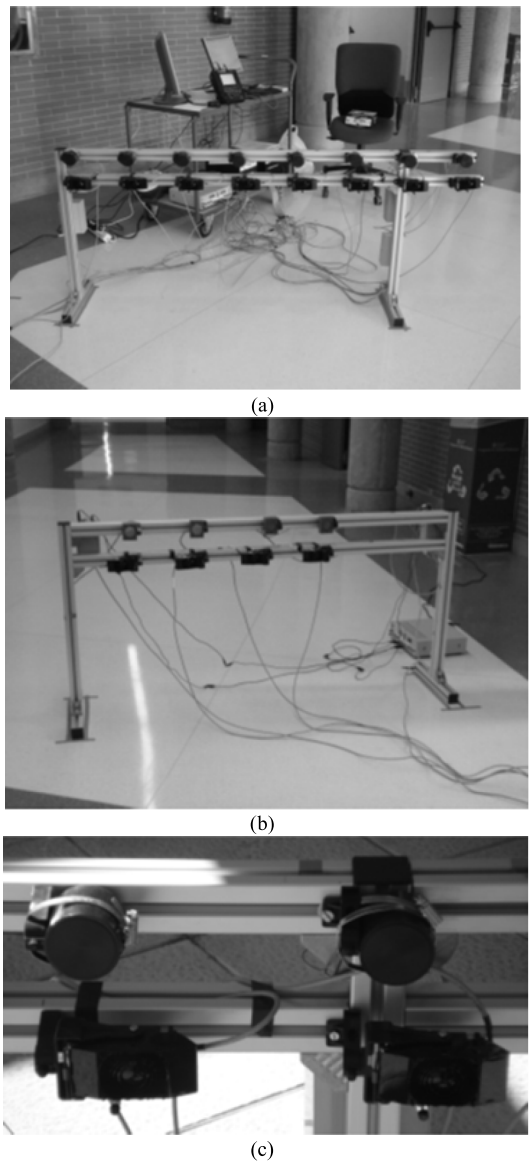


Fig. 19. Block diagram of the correlation proposed for the IR receivers.

sample is binarized by comparing it with the half of the full range FE of the acquisition system. The last $m^{(\text{in}}$ bits obtained by binarizing $r_i^{(\text{in}}[n]$ are stored to determine the position of the pulse by means of a multiplexer. This demodulation implies a significant reduction in the data width for the signal $d_i^{(\text{in}}[n]$, compared with the noncoherent US implementation.

The correlation module is the same for both US and IR signals. Fig. 19 describes the one proposed for IR, where the operation frequencies have been changed. Note that the input buffer is also reduced thanks to the coherent detection.



(a)

(b)

(c)

Fig. 20. (a) Aspect of the US/IR transmitting barrier. (b) Aspect of the US/IR receiving barrier. (c) Details of the US transducer (top) and IR transducer (bottom).

Finally, the ROM memories stored the code $c_j^{(\text{in}}$ transmitted by every emitter j, which is formed by the interleaving of the sequences from four-CSS.

Finally, the peak detector in charge of validating the local maximum values as correct transmissions is the same as the one commented on in Fig. 16, assuming a different analysis window $F^{(\text{in}}$ and threshold $U^{(\text{in}}$.

V. Results

The IR/US multisensory barrier has been implemented according to the geometry shown in Fig. 2. Fig. 20 shows the implemented prototype as well as a detail of the US and IR transducers. The US ones are the model ATK50 by Airmar Technology Corporation [28]. This transducer has a maximum frequency response at 50 kHz, a bandwidth of 4 kHz, and a beamwidth of 10°.

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TABLE I
RESOURCE CONSUMPTION IN THE THREE FPGA-BASED MODULES FOR AN XC3S1200E

Resource type	Module 1 US/IR emitters	Module 2 US receivers	Module 3 IR receivers
Slices	486 (5.6%)	7041 (81.2%)	5938 (68.5%)
BRAMs	4 (14.3%)	16 (57.1%)	16 (57.1%)
IOBs	29 (14.9%)	74 (38.1%)	75 (38.7%)

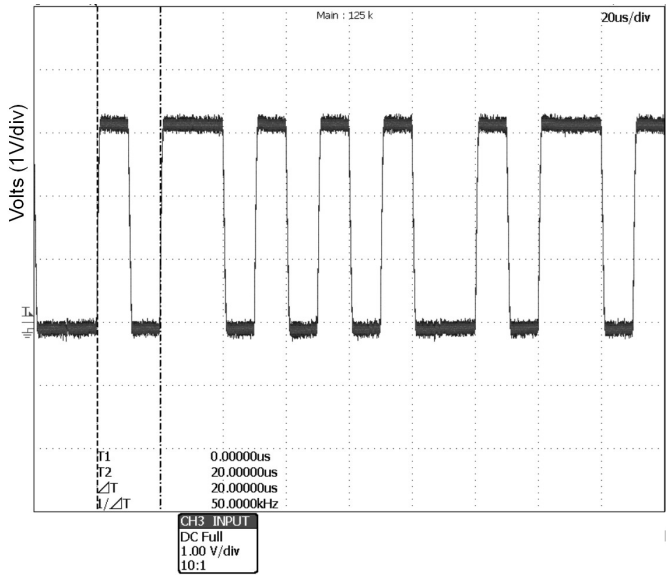


Fig. 21. Details of the signal $e_j^{(\mathrm{us})}[n]$ transmitted by the US emitter j after the BPSK modulation.

The SensComp 6500 ranging module has been modified and used for driving the US transducers [29]. With regard to the IR ones, the UNIRIS system by Sorhea has been used [30]. This transducer presents a maximum of emitting energy at 0°, whereas at the other working angles, 1° and 2°, 74% and 57% over the maximum emitting intensity are obtained, respectively.

The processing and control algorithms explained in the previous sections have been implemented on a Nexys2 board by Digilent Inc. [31]. This is based on a Xilinx XC3S1200E FPGA [32], with D/A and A/D channels based on the DAC121S101 [26] and ADCS7476 [27] devices mentioned before, respectively. Three Nexys2 modules have been used: 1) the first one manages the US and IR emitters, with eight D/A channels; 2) the second one controls the US receivers, with eight A/D channels; and 3) the third one is connected to the IR receivers, also with eight A/D channels. In Table I, the resource consumption of each FPGA device is shown. Note that the devices still have available resources, so the number of US/IR emitters/receivers could be increased in future extensions of the barrier due to its flexibility and modularity, even by replacing the current devices by other larger ones. Furthermore, it is important to take into account that in the same way, the three FPGA devices can also be replaced by only one large enough FPGA, obtaining the same functional behavior.

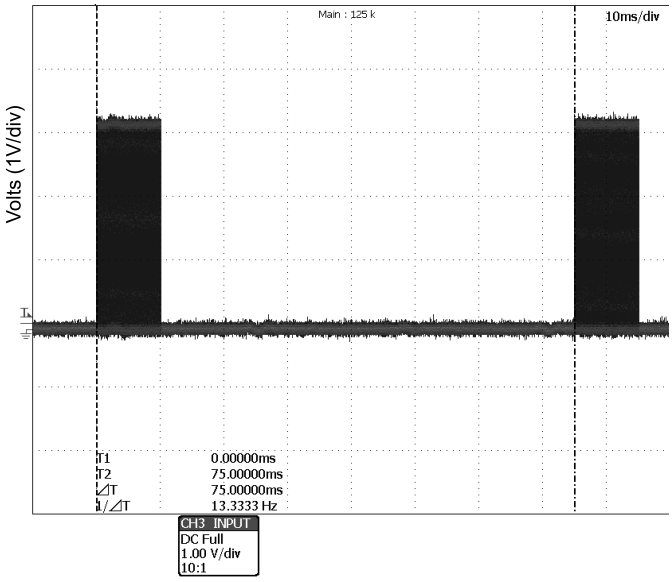


Fig. 22. Signal $e_j^{(\mathrm{us})}[n]$ transmitted by a US transducer j every 75 ms.

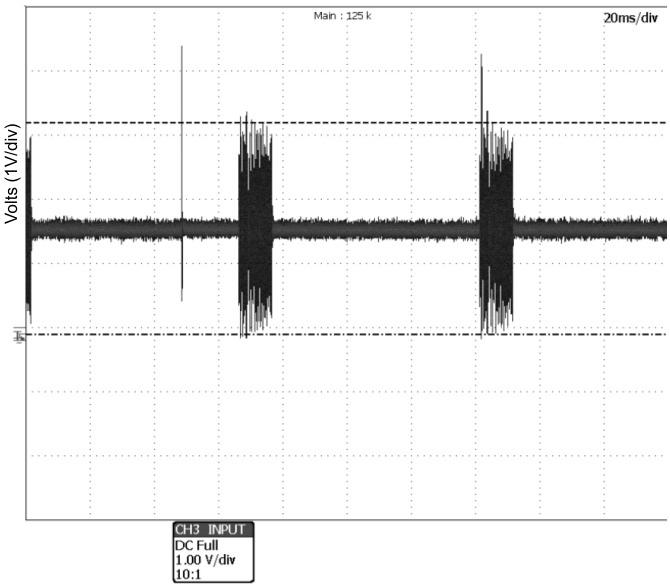


Fig. 23. Signal $r_i^{(\mathrm{us})}[n]$ received by a US transducer i at the input of the ADC.

On the mentioned Nexys2 board, with a clock frequency $f_{\mathrm{CLK}} = 50$ MHz, every new US sample $r_i^{(\mathrm{us})}[n]$ is fully processed in 2.6 μs (in order to determine whether it is a transmission peak), whereas every IR sample $r_i^{(\mathrm{ir})}[n]$ is computed after 5.5 μs. Furthermore, a new emission process is carried out at a scanning frequency of 13 Hz, as has been already explained. These processing times together with the scanning frequency allow meeting the maximum scanning time previously indicated (500 ms according to [8]).

Fig. 21 depicts the emission $e_j^{(\mathrm{us})}[n]$ encoded with a Kasami sequence $c_i^{(\mathrm{us})}$ and BPSK modulated with a carrier period of 20 μs. On the other hand, due to constraints in the driving module of US transducers, an encoded emission is carried out every 75 ms, as can be observed in Fig. 22.

With regard to the reception, the signal $r_i^{(\mathrm{us})}[n]$ captured by the US transducer is processed, so its range is between

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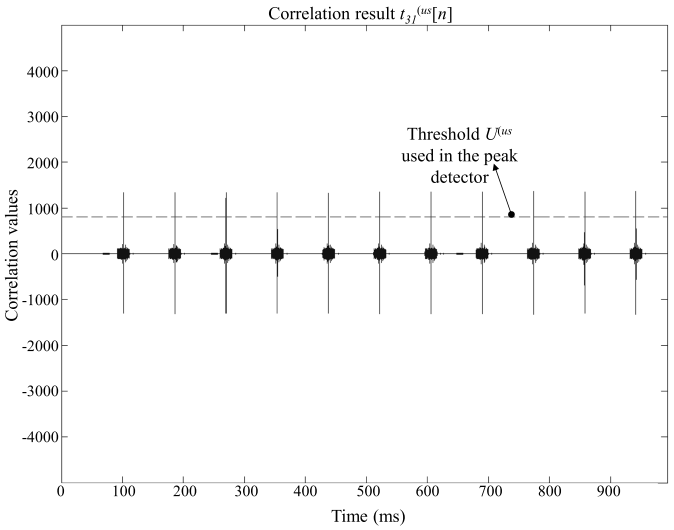


Fig. 24. Example of a correlation signal $t_{31}^{(\mathrm{us}}[n]$.

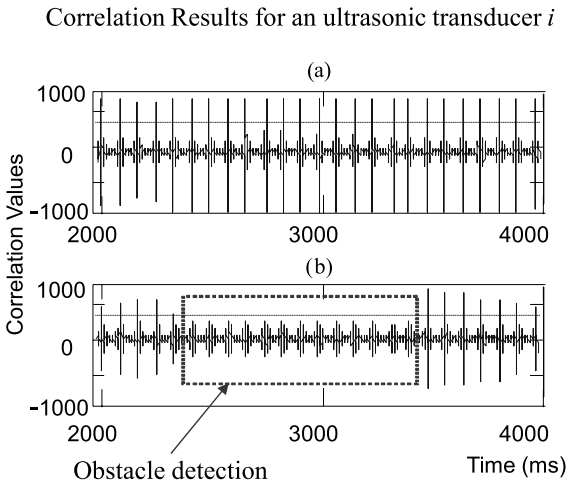


Fig. 25. Correlation signals when an obstacle is detected. (a) $t_{i3}^{(\mathrm{us}}[n]$. (b) $t_{i4}^{(\mathrm{us}}[n]$.

0 V and 3.3 V at the input of the ADC to take advantage of the full scale. Fig. 23 shows an example of a US reception $r_i^{(\mathrm{us}}[n]$, where the transmissions are captured every 75 ms. This reception is demodulated and correlated in order to determine the correlation peaks. These maximum values are compared with a threshold $U^{(\mathrm{us}}$ to validate the detection. In Fig. 24, the correlation output $t_{ij}^{(\mathrm{us}}[n]$ with the corresponding peaks can be detected, as well as the threshold $U^{(\mathrm{us}}$ can be used. The value of this threshold $U^{(\mathrm{us}}$ should be selected depending on the environmental conditions.

Furthermore, Fig. 25 shows two of the four correlation functions $t_{ij}^{(\mathrm{us}}[n]$ for a receiver i. It can be observed how the link with the emitter $j = 4$ is interrupted by the presence of an obstacle. This implies a reduction below the detection threshold $U^{(\mathrm{us}}$ of the correlation maximum values, so the obstacle presence can be confirmed. In this case, the obstacle is a box with the minimum size to be detected, $50 \times 50 \times 50$ cm^3, but only interrupting one of the external links ($j = 4$).

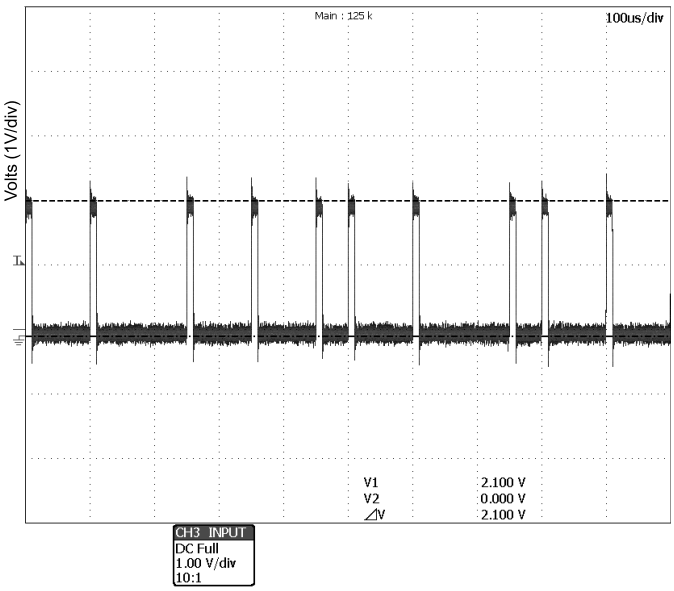


Fig. 26. Encoded signal $e_j^{(\mathrm{in}}[n]$ to be emitted by an IR transducer j.

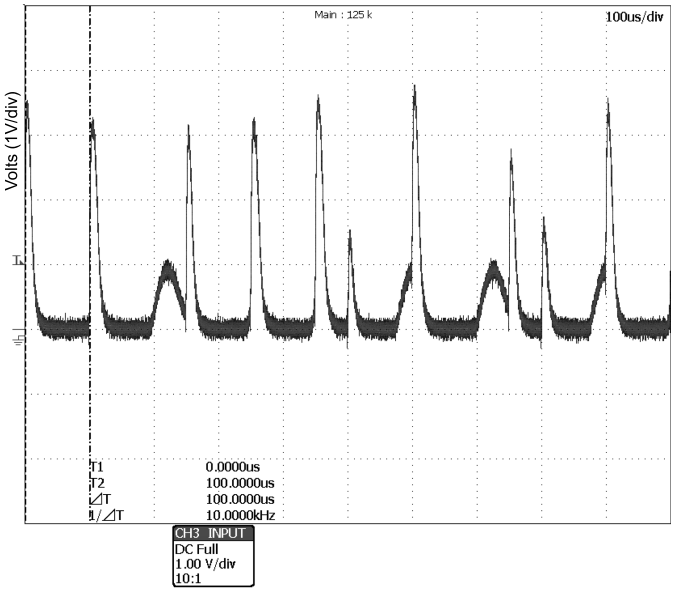


Fig. 27. Signal received $r_i^{(\mathrm{in}}[n]$ for a generic IR link.

With regard to the IR barrier, similar tests have been carried out to those shown for the US one previously. In Fig. 26, it can be observed the encoded emission $e_j^{(\mathrm{in}}[n]$ that drives an IR emitter j, where the width of pulses is 10 μs and the carrier period is 100 μs, according to the proposed PPM modulation. Figs. 27 and 28 show the received signals $r_i^{(\mathrm{in}}[n]$ after the IR link. In the first case, a generic reception is shown, whereas a detail of the reception of a PPM modulation symbol is depicted in Fig. 28. Finally, in Fig. 29, a real example of a correlation result $t_{ij}^{(\mathrm{in}}[n]$ can be observed, where it is possible to check the effect of the appearance of an obstacle during normal operation.

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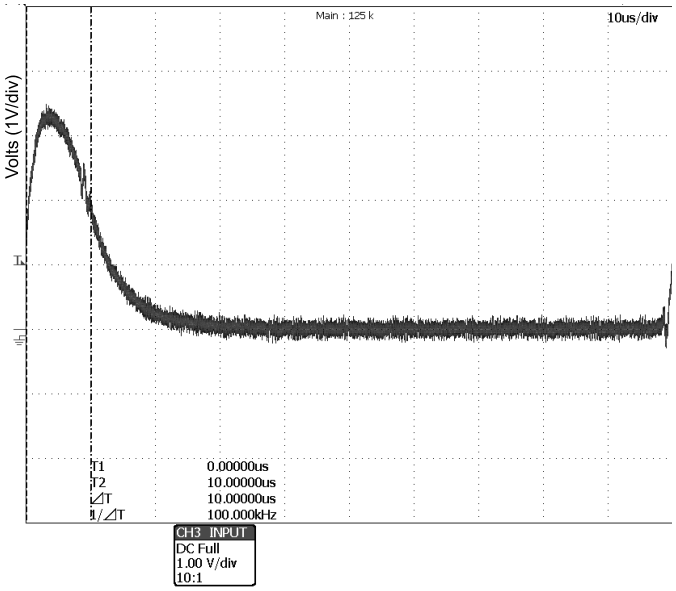


Fig. 28. Details of the reception of the PPM modulation symbol.

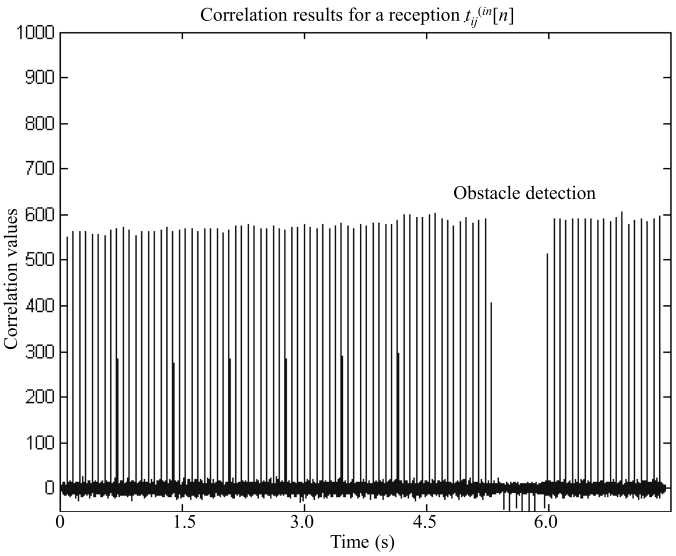


Fig. 29. Correlation result $t_{ji}^{(\text{in}}[n]$ with the detection of an obstacle.

VI. Conclusion

This paper describes the development and implementation of a multisensory safety barrier based on IR and US sensors for railway environments. It proposes the design of an emitter barrier and a receiver one, each placed at one side of the railway. Several links are defined between emitters and receivers, based on US and IR transducers in order to detect the possible presence of obstacles. This allows the security of the facilities to be increased, especially in some crucial places such as bridges or tunnels. The sensory signals have been encoded to allow simultaneous emissions and receptions avoiding crosstalk interferences. This makes possible even to approximately compute the position of the obstacle inside the scanned area. The signal encoding also improves the noise immunity of the system and the global performances. Finally, all the proposed processing has been implemented in a prototype, based on low-cost FPGA devices, providing real-time successful experimental tests.

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